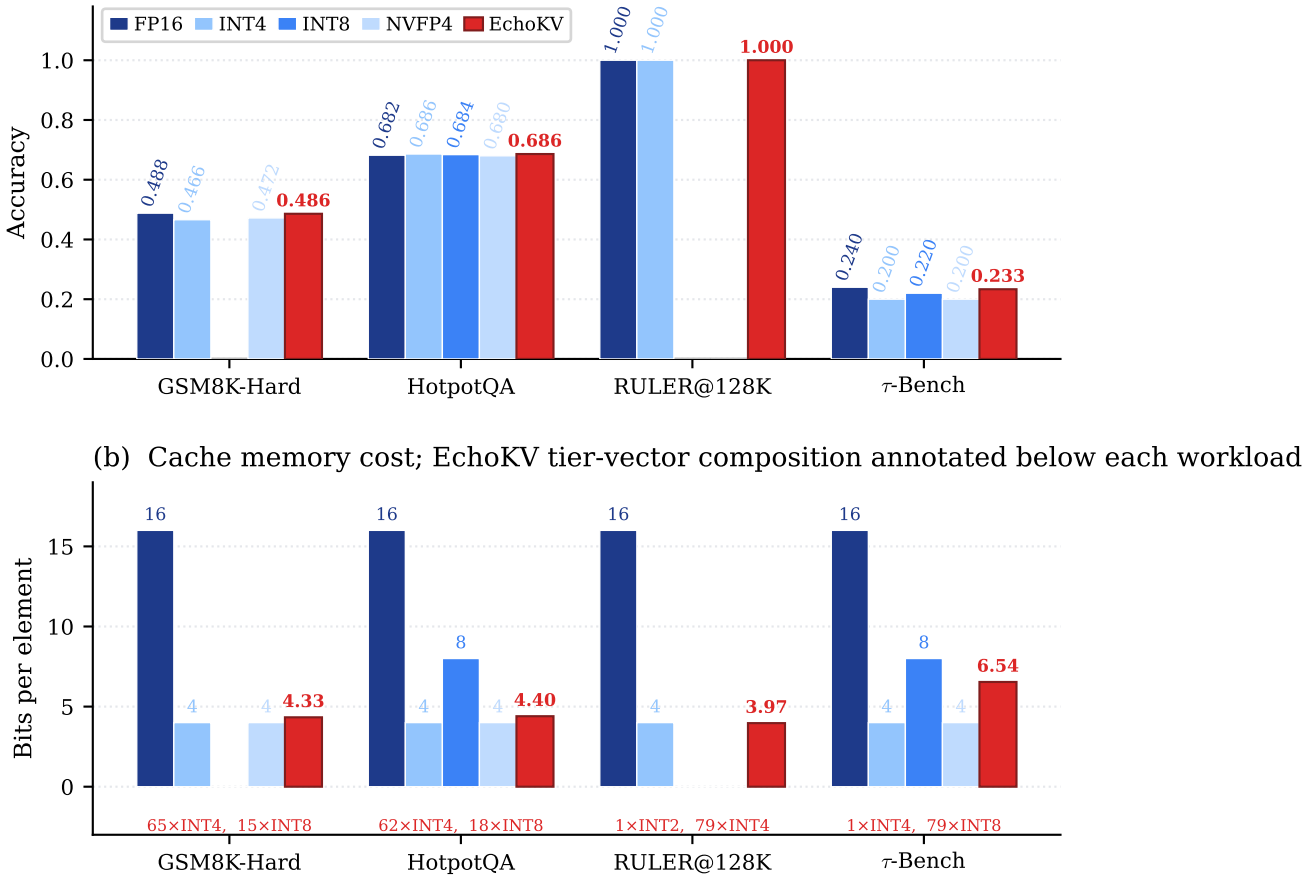
(a) Task accuracy (paired protocol; same prompts across cells)
FP16
INT4
INT8
NVFP4
EchoKV
Accuracy
0.0
0.2
0.4
0.6
0.8
1.0
0.488
0.466
0.472
0.486
0.682
0.686
0.684
0.680
0.686
1.000
1.000
1.000
0.240
0.200
0.220
0.200
0.233
GSM8K-Hard
HotpotQA
RULER@128K
τ-Bench
(b) Cache memory cost; EchoKV tier-vector composition annotated below each workload
Bits per element
0
5
10
15
16
4
4
4.33
16
4
8
4
4.40
16
4
3.97
16
4
8
4
6.54
65×INT4, 15×INT8
62×INT4, 18×INT8
1×INT2, 79×INT4
1×INT4, 79×INT8
GSM8K-Hard
HotpotQA
RULER@128K
τ-Bench